FIRST-ORDER CORRECTION TO THE HALL CONDUCTIVITY OF WEAKLY INTERACTING MASSIVE DIRAC FERMIONS

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Abstract

The aim of this project was to calculate first-order corrections to the anomalous Hall conductivity of massive Dirac fermions, due to short-range interactions. This was performed in the finite temperature formalism, using Matsubra Green's functions as fundamental objects in the context of diagrammatic perturbation theory. For the non-interacting case, the behaviour of the Hall conductivity with varying temperature and chemical potential was obtained. This result proved to be consistent both with the work from the previous semester and the literature. For the first-order corrections, the conservation of the number of particles was imposed, in order to amend for the unphysical behaviour introduced by truncating the perturbation series. The result obtained suggests that this first-order contribution vanishes in the limit of zero temperature and zero chemical potential, as postulated by Coleman, Hill (1985).

1 Introduction

Many-body systems are of paramount interest in theoretical condensed matter, since they enable us to study the behaviour of a macroscopic number of interacting variables (Kotov et al., 2012), which often give rise to fascinating physical phenomena. Graphene is among the materials which serve as the playground for modelling some of these ideas, as it hosts a unique electron system from the point of view of interactions (Neto et al., 2009).

The project carried out last semester was dedicated to bridging the gap between topology and the quantisation of the Hall conductivity of massive Dirac fermions. These are found as the low-energy excitations of two-dimensional hexagonal lattices with broken inversion symmetry, including graphene with a finite gap. Further on, this thesis is focused on calculating its first-order corrections due to short-range Coulomb interactions modelled by a contact potential, which we generalise for finite temperatures.

After introducing the basis of Matsubara Green's functions, we use diagrammatic perturbation theory to obtain the first-order vertex and self-energy corrections to the current-current correlation function in imaginary time. Once analytically continuing back to real time, we retrieve the causal response function, which we then use to compute the effect of the interactions, by applying the Kubo formula. Since we are employing a perturbative approach, our results are only relevant for weakly interacting massive Dirac fermions.

2 Massive Dirac fermions

The goal of this project is to study the effects of interactions on the Hall conductivity of two-dimensional massive Dirac fermions, which are particles with non-zero mass described by the Dirac equation. In graphene, although the energy spectrum near the two Dirac points is gapless, a gap can be opened by breaking the inversion symmetry of the system, which leads to the fermions acquiring a mass (Girvin, Yang, 2019).

This can be achieved experimentally by placing it on a substrate of born nitride (BN)or silicon carbide (SiC), which produces an inhomogeneous electrostatic potential that breaks the sublattice symmetry (Guan et al., 2017; Sahu, Rout, 2017). Theoretically, massive Dirac fermions can be found in the Haldane model (Haldane, 1988), where the sublattice symmetry is broken by introducing a next-nearest-neighbour hopping term in the tight-binding method.

Restricting ourselves to the K-valley alone, the non-interacting Hamiltonian describing such a system and its energy dispersion are given by:

$$\hat{H}_0(\boldsymbol{k}) = \hbar v_{\mathrm{F}}\, \boldsymbol{k}\cdot\boldsymbol{\sigma} + \Delta\sigma_z, \tag{2.1}$$

$$\varepsilon^{\pm}_{\boldsymbol{k}} = \pm\sqrt{(\hbar v_{\mathrm{F}})^2 k^2 + \Delta^2} = \pm\,\varepsilon_{\boldsymbol{k}}, \tag{2.2}$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$, since we are dealing with two-dimensional excitations, Δ is half the energy gap, v_{F} is the Fermi velocity and "$\pm$" denotes the band index.

2.1 Review of topological aspects

Similar to the Hall effect which gives rise to a transverse current when a magnetic field is applied to a metal, Karplus, Luttinger (1954) showed that when an external electric field is applied instead, electrons acquire an additional contribution to their group velocity. Despite the fact that this distinctive effect has been known ever since Edwin H. Hall's momentous discovery in 1879 (Hall, others, 1879), it wasn't until 1984 when the concept of Berry phase (Berry, 1984) finally shed light on the underlying mechanism behind it, by relating its origin to the topology of the band structure.

In Dumitriu-Iovanescu (2021), Chapter 2 was dedicated to introducing the concepts of Berry phase, connection and curvature, where we also examined the physical consequences they have in a cyclic adiabatic process in the parameter space. Using Chern's theorem, we related the integral over the Brillouin zone of the Berry curvature to a topological constant of the manifold, the Chern number. The Kubo formula, which we are going to revisit next, enabled us to express the Hall conductivity in terms of the transverse current-current response function, which we also proved to be correlated to the Chern number. This demonstrated that the transverse conductivity is a topological property of the system, dictated by the energy dispersion of the band structure.

The latter motivated our interest in massive Dirac fermions, since the topology of a gapped energy spectrum enables us to separate the Hall conductivity into a topological part (coming from the filled band) and a non-topological one (due to the doping). We also discussed the symmetry considerations needed to ensure a non-vanishing Hall conductivity, and proved that if either inversion or time-reversal symmetry are present, the net effect from the two Dirac points on σ_{H} is zero.

2.2 The Kubo formula

The main approach used to perform ab initio calculations of the Hall conductivity is the Kubo formalism (Kubo, 1957). It relates the macroscopic electrical conductivity to the causal current-current response function via the formula:

$$\sigma_{\alpha\beta}(\omega) = \underbrace{\frac{ie^2}{\omega}\frac{n}{m}\delta_{\alpha\beta}} + \frac{ie^2}{\omega}\chi_{j_{p\alpha}j_{p\beta}}(\boldsymbol{q}=0,\omega), \tag{2.3}$$

where α, β give the directions of the two current operators $\hat{j_\alpha}$ and $\hat{j_\beta}$ and the "p" label indicates that we are only taking the paramagnetic component of the current density.

The Hall conductivity is defined as the DC limit $\omega \to 0$ of the transverse electrical conductivity: $\sigma_{\text{H}} = \lim_{\omega\to 0}\sigma_{xy}(\omega)$ (Murayama, 2008), which ensures that the marked term in the equation above vanishes. The key advantage of using this method is that it enables us to account for the effects of interactions, since we can calculate the causal response function by means of Feynman diagrammatics.

2.3 Interactions

One of the most interesting aspects found when studying the properties of graphene is that its energy dispersion mimics the physics of quantum electrodynamics, as we saw when we introduced Dirac fermions. Knowing that unusual properties from QED are also

present in graphene, but at much smaller speeds, prompts us to investigate the effects of electron-electron interaction. For typical values of these speeds of order $v_{\text{F}} \sim 10^6\,\text{m/s}$, we can define a fine-structure constant for graphene $\alpha = e^2/(4\pi\varepsilon_0\hbar v_{\text{F}})$, which is found in Gan et al. (2016) to fall in the range $0.25 - 0.35$. Since this is much higher than $\alpha_{\text{QED}} = 1/137$, we deduce that interactions are much stronger in graphene than what we expect in QED, such that perturbative approaches need to be treated carefully.

The most common type of electron-electron interaction to consider is the Coulomb one, which has different forms in two and three dimensions, depending on how we model our system[1]. However, since both these forms are inversely proportional to the momentum carried, setting $\boldsymbol{q} \to 0$ gives rise to various IR divergences, which make the calculation extremely challenging

A way around this is to only consider short-range Coulomb interactions and approximate them by a contact potential V_c, which does not depend on the momentum carried:

$$V_c = \lambda \frac{e^2}{\varepsilon_0 \Lambda}, \tag{2.4}$$

where λ is the strength of the interaction, ε_0 the vacuum permittivity and Λ the natural cut-off in momentum, which is inversely proportional to the lattice spacing (Neto et al., 2009).

3 Finite temperature formalism

In addition to the zero temperature linear response theory discussed in the previous report (Dumitriu-Iovanescu, 2021), we are now interested in extending our analysis to phenomena closer to the experimental terrain, where the effects of temperature are often meaningful. For this reason, this chapter is dedicated to introducing a finite temperature formalism and to demonstrating how we can generalise our previous results by analysing the evolution of the Hall conductivity corrections with temperature.

By temperature dependence we are referring to the fact that our electrons are in fact interacting with a bath of (other) particles, which have a mean energy and therefore a finite temperature (Mahan, 1990). The exact state of this bath is irrelevant for us, as its only in-print on our system consists on electrons having non-zero probabilities of occupying higher energy states, such that from here on we need Fermi-Dirac statistics to characterise our massive Dirac fermions.

The first step in building a finite temperature formalism is to do a mapping to imaginary time, which although not intuitive, is a widely used mathematical trick, particularly useful when dealing with perturbation theory (Bruus, Flensberg, 2004). This will enable us to relate the contributions from interactions to correlation functions which are calculated from Feynman diagrams, while for the end result we are required to analytically continue back to real time in order to determine the Hall conductivity.

As already discussed in Chapter 3.1. from Dumitriu-Iovanescu (2021), the response of the observable $\hat{A}$ to a perturbation that coupled to the observable $\hat{B}$ at an earlier time is quantified by the retarded linear response function $\chi^{\text{R}}_{\text{AB}}$[2]:

$$\chi^{\text{R}}_{\text{AB}}(t) = -\frac{i}{\hbar}\,\theta(t)\,\langle[\hat{A}(t), \hat{B}(0)]_{\text{B/F}}\rangle\,. \tag{3.1}$$

[1] We can either solve the Poisson equation in 3D and set the third component to zero in the final result, or solve the Poisson equation directly in 2D.

[2] We use $[\ldots]_{\text{B/F}}$ to denote a commutator for bosonic operators and an anticommutator for fermionic ones.

The first step in extending our formalism from zero to finite temperature is therefore to switch from the expectation value in the ground state to the grand-canonical ensemble average at some temperature T and chemical potential μ.

For an observable $\hat{O}$, its expectation value in the Hamiltonian basis can be written as:

$$\begin{aligned}\langle\hat{O}\rangle &= \sum_n P_n \langle\psi_n|\,\hat{O}\,|\psi_n\rangle = \sum_n \frac{1}{\mathcal{Z}} e^{-\beta(E_n-\mu n)} \langle\psi_n|\,\hat{O}\,|\psi_n\rangle = \sum_n \frac{1}{\mathcal{Z}} \langle\psi_n|\, e^{-\beta(\hat{H}-\mu\hat{N})}\hat{O}\,|\psi_n\rangle \\ &= \sum_n \langle\psi_n|\left(\frac{1}{\mathcal{Z}} e^{-\beta(\hat{H}-\mu\hat{N})}\right)\hat{O}\,|\psi_n\rangle = \sum_n \langle\psi_n|\,\hat{\rho}\,\hat{O}\,|\psi_n\rangle = \mathrm{Tr}\left\{\hat{\rho}\,\hat{O}\right\}, \end{aligned} \tag{3.2}$$

where $\hat{H}$ is the Hamiltonian, $\hat{N}$ the number operator, $\hat{\rho}$ the grand-canonical density matrix and $\mathcal{Z}$ the partition function. Since we are always working in the grand canonical ensemble and $\{|\psi_n\rangle\}$ are simultaneous eigenstates of both $\hat{H}$ and $\hat{N}$, we are allowed to shorten our notation by simply using $\hat{H}$ instead of $\hat{H}-\mu\hat{N}$.

Although at this point we could argue that we can employ a similar formalism as we did at zero-temperature by relating the response function $\chi^{\mathrm{R}}_{\mathrm{AB}}(t)$ to the time-ordered correlation function $\chi^{T}_{\mathrm{AB}}(t) = -i\,\langle T[\hat{A}(t),\hat{B}(0)]\rangle$, we encounter a major issue: dealing with interactions by means of real-time correlations functions is not amendable at finite temperature by perturbation theory (Jishi, 2013). Consequently, a different approach is required, which prompts us to define a new object:

$$\chi^{\mathcal{T}}_{\mathrm{AB}}(\tau) = -\frac{1}{\hbar}\,\mathrm{Tr}\left\{\hat{\rho}\mathcal{T}[\hat{A}(\tau)\hat{B}(0)]\right\}, \tag{3.3}$$

known as the temperature or time-ordered correlation function in imaginary time (Giuliani, Vignale, 2005). Note that the time dependence of the operators suggests they are evaluated in the Heisenberg picture, which is a detail we are going to explore in the following section.

3.1 Imaginary time

Before moving on to the technicalities of this formalism, it is necessary to provide a more intuitive picture revealing the reasoning behind the apparently unphysical imaginary time, which was found in Bruus, Flensberg (2004). Making a parallel between the grand-canonical density matrix $\hat{\rho} = 1/\mathcal{Z}e^{-\beta\hat{H}}$ and the time evolution operator $\hat{U}(t) = e^{-i\hat{H}t}$, we notice that they are both exponential functions of the Hamiltonian obeying similar equations: $i\partial_t\hat{U} = \hat{H}\hat{U}$ (the Schrödinger equation) and $\partial_\beta\hat{\rho} = -\hat{H}\hat{\rho}$ (the Bloch equation). This comparison suggests that by replacing the time argument with an imaginary variable $t \to -i\tau$, where τ is real, we can obtain both $\hat{\rho}$ and $\hat{U}$ in our interaction power series[3].

In order to build our way up to calculating the effects of interactions by means of Feynman diagrammatics, we need to express the correlation function in terms of the scattering matrix, whose power series bridges the gap between our theoretical formalism and the graphical representation of the physical process, using Feynman diagrams.

Similar to our approach in quantum mechanics, we can define Heisenberg (3.4a) and Dirac (3.4b) picture operators in imaginary time:

$$\hat{A}_H(\tau) = e^{\frac{\tau}{\hbar}\hat{H}}\,\hat{A}\,e^{-\frac{\tau}{\hbar}\hat{H}}, \tag{3.4a}$$

$$\hat{A}_I(\tau) = e^{\frac{\tau}{\hbar}\hat{H}_0}\,\hat{A}\,e^{-\frac{\tau}{\hbar}\hat{H}_0}, \tag{3.4b}$$

[3]By this we mean a power series in the component of the full Hamiltonian $\hat{H}$ which contains the interaction.

where $\hat{H} = \hat{H}_0 + \hat{H}_I$ is the full Hamiltonian, which includes both the non-interacting part $\hat{H}_0$ and the perturbative one due to the interaction $\hat{H}_I$. In order to distinguish the real time operators from the imaginary time ones, our convention from here on is to use Roman (t) and Greek (τ) letters for each of them, respectively.

We can relate the two equations in (3.4) in the form $\hat{A}_H(\tau) = \hat{U}_I(0,\tau)\hat{A}_I(\tau)\hat{U}_I(\tau,0)$ by introducing the time-evolution operator $\hat{U}_I$ in the interaction picture:

$$\hat{U}_I(\tau,\tau_0) = e^{\hat{H}_0\frac{\tau}{\hbar}}\, e^{-\hat{H}\frac{\tau-\tau_0}{\hbar}}\, e^{-\hat{H}_0\frac{\tau_0}{\hbar}}. \tag{3.5}$$

In the special case when $\tau = \hbar\beta$, $\tau_0 = 0$, it can be written as:

$$\hat{U}_I(\hbar\beta,0) = e^{\beta\hat{H}_0}\, e^{-\beta\hat{H}} = \mathcal{T}\exp\left(-\frac{1}{\hbar}\int_0^{\hbar\beta}\hat{H}_I(\tau')d\tau'\right) \equiv \hat{S}, \tag{3.6}$$

where the second identity is assumed without proof (see chapter 4.2. from Peskin (2018) for more details). The only difference between this scattering matrix and the one we are familiar with from QFT is the upper limit which instead of infinity, only extends up to $\hbar\beta$, a restriction which we are going to clarify in the next part.

3.2 Periodicity considerations

Looking at the operators defining our temperature correlation function in (3.3), we should point out that the equation does not make sense mathematically for all values of τ. Since we "absorbed" the number of particles operator into our Hamiltonian, $e^{\pm\hat{H}\tau/\hbar}$ can have arbitrarily large eigenvalues, which in order to ensure the convergence, should always be counterbalanced by the exponentially small eigenvalues of the density matrix operator $e^{-\beta\hat{H}}$. This constrains τ to the range $-\hbar\beta < \tau < \hbar\beta$ (Giuliani, Vignale, 2005).

For τ within the restriction $0 > \tau > -\hbar\beta$, we can calculate $\chi^{\mathcal{T}}_{\mathrm{AB}}(\tau+\hbar\beta)$:

$$\begin{aligned}
\hbar\,\chi^{\mathcal{T}}_{AB}(\tau+\hbar\beta) &= -\operatorname{Tr}\{\hat{\rho}\mathcal{T}[\hat{A}(\tau+\hbar\beta)\hat{B}]\} \\
&= -(-1)^f\mathcal{Z}^{-1}\operatorname{Tr}\left\{e^{-\beta\hat{H}}e^{\hat{H}(\tau/\hbar+\beta)}\hat{A}e^{-\hat{H}(\tau/\hbar+\beta)}\hat{B}\right\} \\
&= -(-1)^f\mathcal{Z}^{-1}\operatorname{Tr}\left\{\cancel{e^{-\beta\hat{H}}e^{\beta\hat{H}}}e^{\hat{H}\tau/\hbar}\hat{A}e^{-\hat{H}\tau/\hbar}e^{-\beta\hat{H}}\hat{B}\right\} \\
&= -(-1)^f\mathcal{Z}^{-1}\operatorname{Tr}\left\{e^{-\beta\hat{H}}\hat{B}e^{\hat{H}\tau/\hbar}\hat{A}e^{-\hat{H}\tau/\hbar}\right\} \\
&= -(-1)^f\mathcal{Z}^{-1}\operatorname{Tr}\{\hat{\rho}\mathcal{T}[A(\tau)\hat{B}]\} \\
&= (-1)^f\hbar\,\chi^{\mathcal{T}}_{AB}(\tau),
\end{aligned} \tag{3.7}$$

where we used the fact that for $\tau+\hbar\beta > 0$, $\mathcal{T}[\hat{A}(\tau+\hbar\beta)\hat{B}] = \hat{A}(\tau+\hbar\beta)\hat{B}$, as well as the property that the trace is invariant under cyclic permutations. Note that f represents the number of interchanges of fermion operators, which can appear in the time ordering when $\hat{A}$ and $\hat{B}$ are fermionic. A similar proof for $\hbar\beta > \tau > 0$ can be found in Chapter 6.4. from Giuliani, Vignale (2005), such that we can combine the two into a single equation:

$$\chi^{\mathcal{T}}_{AB}(\tau) = (-1)^f\chi^{\mathcal{T}}_{AB}(\tau - \hbar\beta\,\mathrm{sign}(\tau)). \tag{3.8}$$

From this we draw the conclusion that depending on the parity of f, the correlation function is either periodic or antiperiodic in τ, with period $\hbar\beta$.

3.3 Matsubara frequencies

Knowing that the temperature correlation function has both a periodic and an antiperiodic part naturally leads to writing it as a Fourier series:

$$\chi^{\mathcal{T}}_{AB}(\tau) = \frac{1}{\hbar\beta}\sum_n \chi^{\mathcal{T}}_{AB}\left(i\omega_n\right) e^{-i\omega_n\tau}, \tag{3.9}$$

where ω_n are called Matsubara frecquencies (Giuliani, Vignale, 2005). When ensuring that Equation (3.8) is satisfied in the Fourier domain, we arrive to the conclusion that these frequencies can only have two forms, depending on the parity of f, which are given by:

$$\omega_n = \frac{2n}{\hbar\beta}\pi \qquad \text{for even } f\text{(bosonic frequencies)}, \tag{3.10a}$$

$$\omega_n = \frac{2n+1}{\hbar\beta}\pi \qquad \text{for odd } f\text{(fermionic frequencies)}, \tag{3.10b}$$

The importance of the Matsubara frecquencies will become more evident later on, after we introduce the Matsubara Green's functions. Additionally, the Fourier coefficients

$$\chi^{\mathcal{T}}_{AB}\left(i\omega_n\right) = \int_0^{\hbar\beta} \chi^{\mathcal{T}}_{AB}(\tau) e^{i\omega_n\tau} d\tau \tag{3.11}$$

are crucial when it comes to calculating the transform of the retarded response function by means of analytic continuation, which will also be revised by the end of this chapter.

3.4 Matsubara Green's functions

After building up this formalism, all that is left to do is to write our temperature correlation function in terms of the scattering matrix, whose expansion will allow us to calculate exactly the first-order correction due to interactions.

Making use of our definition (3.6) for the scattering matrix and "converting" the two operators in Equation (3.3) from the Heisenberg to the Dirac picture allows us to write:

$$\chi^{\mathcal{T}}_{AB}(\tau) = -\frac{1}{\hbar}\frac{\mathrm{Tr}\left\{e^{-\beta\hat{H}_0}\mathcal{T}\left[\hat{A}_I(\tau)\hat{B}_I\hat{S}\right]\right\}}{\mathrm{Tr}\left\{e^{-\beta\hat{H}_0}\hat{S}\right\}}, \tag{3.12}$$

where we substituted the expressions for the density matrix $\hat{\rho} = e^{-\beta\hat{H}}/\mathcal{Z}$ and for the partition function $\mathcal{Z} = \mathrm{Tr}\{e^{-\beta\hat{H}}\} = \mathrm{Tr}\{e^{-\beta\hat{H}_0}\hat{U}_I(\hbar\beta, 0)\}$.

Without delving into the details concerning second quantisation (which were discussed in Chapters 3 and 5 from Dumitriu-Iovanescu (2021)), we now introduce the corresponding Green's Function for imaginary time, also called Matsubara Green's Functions[4]:

$$G^{(0)}_{\alpha\beta}(\tau) = -\left\langle \mathcal{T}[\hat{a}_\alpha(\tau)\hat{a}^\dagger_\beta(0)]\right\rangle, \tag{3.13}$$

where $\hat{a}$ and $\hat{a}^\dagger$ are destruction and construction operators, and $\langle\ldots\rangle$ denotes the average over the grand canonical ensemble. The latter is calculated by taking the trace of the product with the matrix density operator $\hat{\rho}$, as previously shown in Equation (3.2).

[4]For simplicity, even though this has two arguments, $\boldsymbol{r}-\boldsymbol{r}'$ and $t-t'$, we omitted writing explicitly $\boldsymbol{r}-\boldsymbol{r}'$ since we are only dealing with translationally invariant systems, and we chose $t'=0$, such that $t-t'=\tau$.

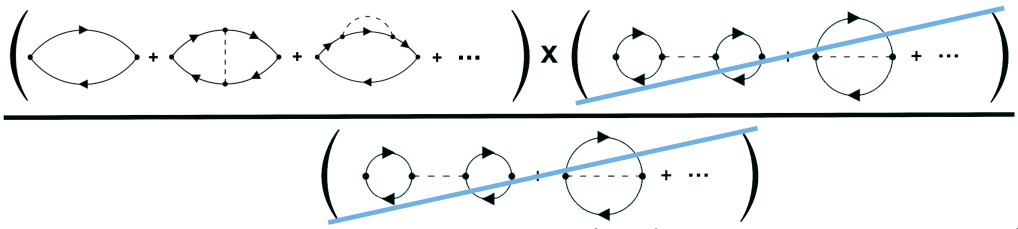


Figure 3.1: Schematic representation of Equation (3.12), inspired from Zagoskin (1998).

Physically, $G^{(0)}_{\alpha\beta}(\tau)$ gives the probability amplitude for a particle in state β added to the non-interacting Fermi sea at "imaginary time" $\tau = 0$ to be found in the state α at some later "time" τ (Giuliani, Vignale, 2005).

For a translationally invariant system, it is particularly useful to go to the Fourier space, which we choose to do in the Hamiltonian basis. Splitting Equation (3.13) into $\tau > 0$ and $\tau < 0$ we get $G^{(0)}_{\alpha\beta}(\tau) = -\delta_{\alpha\beta}\left[\theta(\tau)(1 \pm n_\alpha^{(\mathrm{B/F})}) \mp \theta(-\tau) n_\alpha^{(\mathrm{B/F})}\right] e^{-(\varepsilon_\alpha - \mu)\tau/\hbar}$, where $n_\alpha^{(\mathrm{B/F})}$ are the Bose-Einstein and the Fermi-Dirac distribution functions, respectively. Going to the Fourier domain, the Matsubara Green's function takes the form:

$$G^{(0)}_{\alpha\beta}(i\omega_n) = \frac{\delta_{\alpha\beta}}{i\omega_n - (\varepsilon_\alpha - \mu)/\hbar}, \tag{3.14}$$

where ω_n are the already discussed Matsubara frequencies, given in Equation (3.10).

We can also choose to work in a different basis, in which case the form of the Matsubara Green's function from Equation (3.13) does not prove to be very useful. However, we can use the fact that $G^{(0)}_{\alpha\beta}$ is the Green's function of the non-interacting Schrödinger equation, which in imaginary time is written as[5]:

$$\left[(-\hbar\,\partial_\tau - \hat{H}_0)\, G^{(0)}(\tau)\right]_{\alpha\beta} = \hbar\delta_{\alpha\beta}\,\delta(\tau). \tag{3.15}$$

Although the Green's function has now a matrix and not a scalar form, we can still calculate it using the Fourier transform of the above equation:

$$\left[(i\hbar\omega_n + \mu)\mathbb{1} - \hat{H}_0(\boldsymbol{k})\right] G^{(0)}(\boldsymbol{k}, i\omega_n) = \hbar\,\mathbb{1}, \tag{3.16}$$

which allows us to find $G^{(0)}(\boldsymbol{k}, i\omega_n)$ as the inverse matrix of $(i\hbar\omega_n + \mu)\mathbb{1} - \hat{H}_0(\boldsymbol{k})$. Here, $i\omega_n$ is the Matsubara frequency, $\hat{H}_0(\boldsymbol{k})$ is the Fourier representation of the non-interacting Hamiltonian and μ emerges from the particle number operator $\hat{N}$, which we "absorbed" into the Hamiltonian from the beginning. What paves our way from Green's functions to Feynman diagrams is Wick's theorem (Wick, 1950), which has a similar form at finite temperature to the well-known one from QFT. It states that the time ordering of a string of operators involving more than one particle can be factorised into products of single-particle Green's functions, which allows us to interpret the numerator and denominator in Equation (3.12) by means of Feynman diagrams. The rigorous details preceding this graphical interpretation are identical at finite temperature to the ones derived in the zero temperature formalism and are discussed at length in Peskin (2018), as well as summarised in Section 5.4 from Dumitriu-Iovanescu (2021). By making a perturbative expansion of the scattering matrix in the interaction term, we obtain a schematic interpretation of (3.12), which can be seen in Figure 3.1. This suggests that a straightforward way to evaluate

[5]In addition to this, the spatial arguments we omitted introduce $\delta(\boldsymbol{r} - \boldsymbol{r}')$.

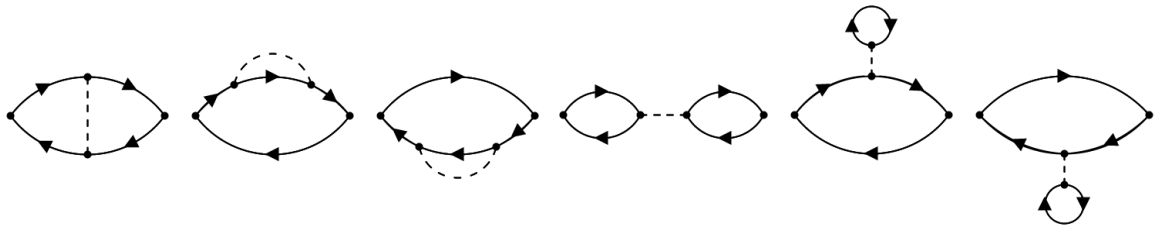


Figure 3.2: Connected, topologically distinct Feynman diagrams of first-order.

the equation is by only considering connected diagrams, since the denominator perfectly cancels the disconnected ones[6].

This allows us to piece together the concepts introduced so far, such that our goal to evaluate the first-order correction to the Hall conductivity comes down to calculating the six connected, topologically distinct[7] diagrams shown in Figure 3.2. The Feynman rules for Matsubara Green's functions are given in Appendix A. The last building block of our formalism is now to develop a method which retrieves the causal response function needed for the Kubo formula from the correlation function, which is the topic of the next section.

3.5 Analytic continuation

We can now calculate the finite temperature correlation function for arbitrary operators $\hat{A}$ and $\hat{B}$ in terms of Matsubara Green's functions, and therefore evaluate the effect that interactions might have. However, we need the response function in real time in order to compute the Hall conductivity, which means that we should focus our attention on retrieving this from the correlation function.

At the core of our formalism lies a remarkable fact, which we will paraphrase here from Bruus, Flensberg (2004), but the complete proof can be found in Appendix B. This states that there exists an analytic function $\chi_{\mathrm{AB}}(z)$, with z a complex, not purely real, frequency argument, which on the imaginary axis is equal to our Matsubara correlation function in frequency domain $\chi_{\mathrm{AB}}(i\omega_n)$, and on the real axis it is the same as the real frequency retarded response function $\chi^{\mathrm{R}}_{\mathrm{AB}}(\omega)$. This property allows us to always determine one from the other by an appropriate analytic continuation. In our case this is obtained by sending $i\omega_n \to \omega + i\eta$, where η is a positive infinitesimal number.

The requirement that η is positive comes from the fact that we want to determine the "retarded" response function, which characterises the present state of the system which has been perturbed at a previous time (Bruus, Flensberg, 2004). Although we usually refer to it simply as the "response function", it is important to emphasise that advanced ones exist as well (and can be recovered from the correlation function by choosing η to be negative), but they do not outline physical effects. This process is illustrated schematically in Appendix B, Figure B.1, both for the retarded and for the advanced response functions. It is crucial to note that performing the analytic continuation by sending $\omega_n \to \omega + i\eta$ can be done only when χ_{AB} is analytic in the upper half plane and is written as a rational function (Bruus, Flensberg, 2004). When the function has branch cuts, there is a different procedure which should be employed, but this detail will not be discussed in this report.

[6]By disconnected diagrams we refer to the ones where the interaction is not linked to the external vertices. Examples of such diagrams are obtained by multiplying the two terms in the product from the nominator.

[7]By topologically distinct diagrams we refer to those one that are not related by a trivial permutation of state labels, as discussed in Section 5.5.1. from Dumitriu-Iovanescu (2021).

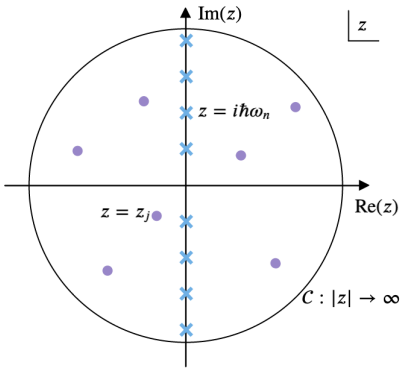


Figure 3.3: Contour $\mathcal{C}:|z|\to\infty$, where the poles on the imaginary axis (blue crosses) correspond to Matsubara frequencies and the scattered ones (purple circles) to those of $g(z)$.

3.6 Matsubara sums

As we will see in the Feynman rules, we are often required to sum over Matsubara frequencies when evaluating correlation functions. We start by defining a general type of sum, which can either be for bosonic or for fermionic frequencies:

$$\mathcal{S}^{\mathrm{B/F}}(\tau) = \frac{1}{\beta}\sum_{i\omega_n} g(i\omega_n)e^{i\omega_n\tau}. \tag{3.17}$$

We can next use the residue theorem to write this sum as the integral over the complex plane, for which we need two functions with poles at the two Matsubara frequencies. These turn out to be precisely the Bose-Einstein and the Fermi-Dirac distribution functions:

$$n_{\mathrm{B}}(z) = \frac{1}{e^{\beta z}-1}, \text{ with poles at } z = i(2k)\pi/\beta, \tag{3.18a}$$

$$n_{\mathrm{F}}(z) = \frac{1}{e^{\beta z}+1}, \text{ with poles at } z = i(2k+1)\pi/\beta, \tag{3.18b}$$

with the residues equal to:

$$\operatorname*{Res}_{z=i\hbar\omega_n}\left[n_{\mathrm{B/F}}(z)\right] = \lim_{z\to i\hbar\omega_n}\frac{(z-i\hbar\omega_n)}{e^{\beta z}\mp 1} = \lim_{\delta\to 0}\frac{\delta}{e^{\hbar\beta(i\omega_n)}e^{\beta\delta}\mp 1} = \pm\frac{1}{\beta}. \tag{3.19}$$

Assuming that $g(z)$ has a number of simple poles[8] at $z = z_j$ and is analytic everywhere else, we can choose a contour which covers the entire complex plane. This contour is shown in Figure 3.3, together with the poles at the Matsubara frequencies (marked by crosses) and the poles of the function $g(z)$ (marked by circles). Since the integrand $n_{\mathrm{B/F}}(z)g(z)e^{z\tau}$ goes to zero on the contour, the contribution from the poles $z = z_j$ and that from the $z = i\hbar\omega_n$, which is $\pm$ our sum given in Equation (3.17), must cancel each other, giving:

$$\mathcal{S}^{\mathrm{B/F}}(\tau) = \mp\sum_j \operatorname*{Res}_{z=z_j}\left[g(z)\right] n_{\mathrm{B/F}}(z_j)e^{z_j\tau}. \tag{3.20}$$

This trick greatly simplifies our initial sum and will be widely used on the sections to follow, when summing over products of Matsubara Green's functions.

[8]This assumption is not arbitrary, as we are often dealing with sums over products of non-interacting Green's function, which only have simple poles.

4 Non-interacting response function

Before diving into the corrections due to interactions, we first need to calculate the non-interacting Hall current-current response function, shown in Figure 4.1(a). As we previously saw in Section 3.4, the form of the Green's function is dependent on the specific choice of basis, which we need to decide upon before evaluating Feynman diagrams.

4.1 Choice of basis

Unlike our approach last semester (Dumitriu-Iovanescu, 2021) when we chose to work in the Hamiltonian basis, we now opted for the σ_z basis, since it seemed to simplify the algebra in the context of the finite temperature formalism[9].

When working in the Hamiltonian basis, the non-interacting Green's function is diagonal, as we can see in Equation (3.14). This means that the state labels, which in our case are the band indices[10], must coincide to the left and to the right of the propagator. Graphically, this translates to assigning a fixed band index to each solid line, which can only change at the vertices, such that we need to sum over all the possible state configurations.

Similar to the phenomenon of spin precession when measuring the spin components of electrons in a uniform magnetic field, the band index of the Dirac fermions also changes along the Matsubara propagator when working in the σ_z basis. This is due to the matrix form on the Green's function, which we can calculate from Equation (3.16):

$$G^{(0)}(\boldsymbol{k}, i\omega_n) = \frac{(i\hbar\omega_n + \mu)\mathbb{1}_2 + \hbar\, v_{\mathrm{F}}\, \boldsymbol{k}\cdot\boldsymbol{\sigma} + \Delta\sigma_z}{(i\hbar\omega_n + \mu)^2 - (\varepsilon_{\boldsymbol{k}})^2}. \tag{4.1}$$

Since the Green's function is now a matrix, it needs two indices to be specified, the same as the vertices, such that including all the contributions from a Feynamn diagram implies summing over pairs of label states. For a closed diagram, this translates to taking the trace of a matrix product, which is particularly useful when the non-interacting Hamiltonian is written in terms of Pauli matrices, which are traceless.

4.2 Non-interacting calculation

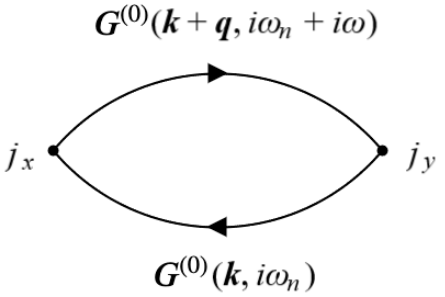
Looking again at Figure 4.1(a), the analytic expression for the operators $\hat{j}_{\mathrm{x/y}} = v_{\mathrm{F}}\,\sigma_{\mathrm{x/y}}$ is obtained using the Feynman rules described in Appendix A:

$$\chi^{\mathcal{T}(0)}_{j_{\mathrm{x}}j_{\mathrm{y}}}(\boldsymbol{q}, i\omega) = -v_{\mathrm{F}}^2 \int \frac{\mathrm{d}^2\boldsymbol{k}}{(2\pi)^2}\frac{1}{\hbar\beta}\sum_{\omega_n}\mathrm{Tr}\Big[\, G^{(0)}(\boldsymbol{k}+\boldsymbol{q}, i\omega_n + i\omega)\,\sigma_x\, G^{(0)}(\boldsymbol{k}, i\omega_n)\,\sigma_y\,\Big], \tag{4.2}$$

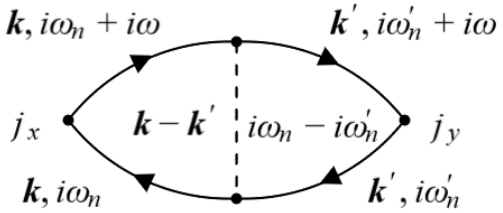
where we summed over the contracted state labels of the two Green's functions and the matrix elements from the two vertices, such that the result is a trace.

[9] One of the major difficulties of working in the Hamiltonian basis was to differentiate single from double poles when considering all the state label configurations (Dumitriu-Iovanescu, 2021), which made it less preferable when repeatedly summing over Matsubra frequencies.

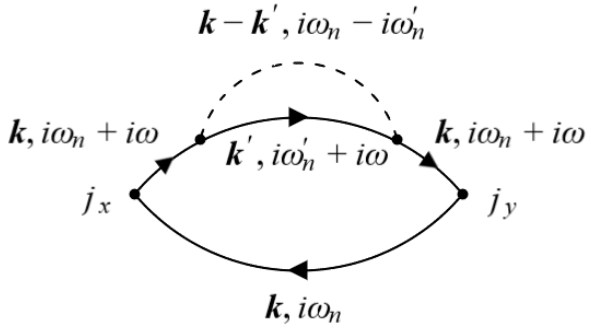
[10] We label the band indices by "+" for the upper band and by "-" for the lower band.



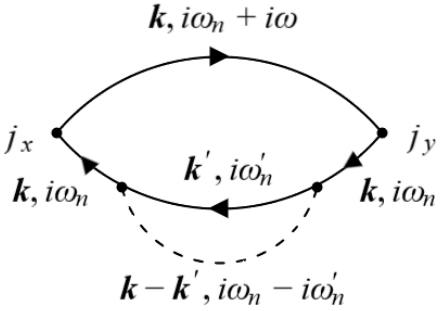
(a) Zero-th order diagram



(b) Exchange diagram



(c) Self-energy 1 diagram



(d) Self-energy 2 diagram

Figure 4.1: Feynman diagrams for the non-interacting transverse current-current response function (on the top left) and the first-order ones that give a correction to it - exchange interaction (on the top right) and the two self energies (bottom).

Using the Green's function from Equation (4.1), we can calculate the numerator of the product inside the trace. After collecting the terms in $\mathbb{1}_2$, σ_x, σ_y, σ_z and using the fact that the trace is a linear mapping[11], only the ones multiplying $\mathbb{1}_2$ remain, as the Pauli matrices are traceless. Once performing the angular integration, the linear terms in $k_x = k\cos\phi$ and $k_y = k\sin\phi$ vanish because of the periodicity, such that we are left with $\mathrm{Tr}[\ldots] = 4\pi\hbar\omega\Delta$, where 2π comes from the integral over $d\phi$ and a factor of 2 from $\mathrm{Tr}[\mathbb{1}_2] = 2$.

Since the numerator is a constant, we now focus on the denominator which has poles at[12] $z = \pm\varepsilon_{\boldsymbol{k}}$ and $z = \pm\varepsilon_{\boldsymbol{k}} - i\hbar\omega$, and perform the Matsubara sum discussed in Section 3.6. Putting everything together we obtain the zero-th order response function:

$$\chi^{\mathcal{T}(0)}_{j_x j_y}(\boldsymbol{q}=\boldsymbol{0}, i\omega) = \frac{v_{\mathrm{F}}^2\,\hbar\omega\,\Delta}{\pi}\int_0^{\Lambda}\frac{\mathrm{d}k\,k}{\varepsilon_{\boldsymbol{k}}}\left(\frac{n^+ - n^-}{4\varepsilon_{\boldsymbol{k}}^2 + (\hbar\omega)^2}\right), \tag{4.3}$$

where Λ is the cut-off in k used to avoid ultraviolet divergences. Also, for the occupation functions of the upper and lower bands, we simplified the notation by $n_{\mathrm{F}}[\pm\varepsilon_{\boldsymbol{k}} - \mu] \to n^{\pm}$.

After analytically continuing $\omega \to -i\omega + \eta$ and using the Kubo formula (2.3), we calculate the Hall conductivity from the expansion of the response function in small ω:

$$\sigma_{\mathrm{H}}(T,\mu) = \frac{e^2}{2h}\frac{1}{T}\int\frac{\mathrm{d}x}{x^2}(n^+ - n^-), \tag{4.4}$$

where we chose to work with adimensional quantities $x = \beta\varepsilon_{\boldsymbol{k}}$ and $T = 1/(\beta\Delta)$, which is

[11] This means that the trace of a sum is equal to the sum of the traces of the individual terms.

[12] The presence of $i\hbar\omega$ in the poles and therefore in the occupation function can be ignored, as the external frequency is bosonic and therefore $n_{\mathrm{F}}(\pm\varepsilon_{\boldsymbol{k}} - i\hbar\omega) = \left(e^{\pm\beta\varepsilon_{\boldsymbol{k}}}e^{\beta i\hbar\omega} + 1\right)^{-1} = \left(e^{\pm\beta\varepsilon_{\boldsymbol{k}}} + 1\right)^{-1} = n_{\mathrm{F}}(\pm\varepsilon_{\boldsymbol{k}})$.

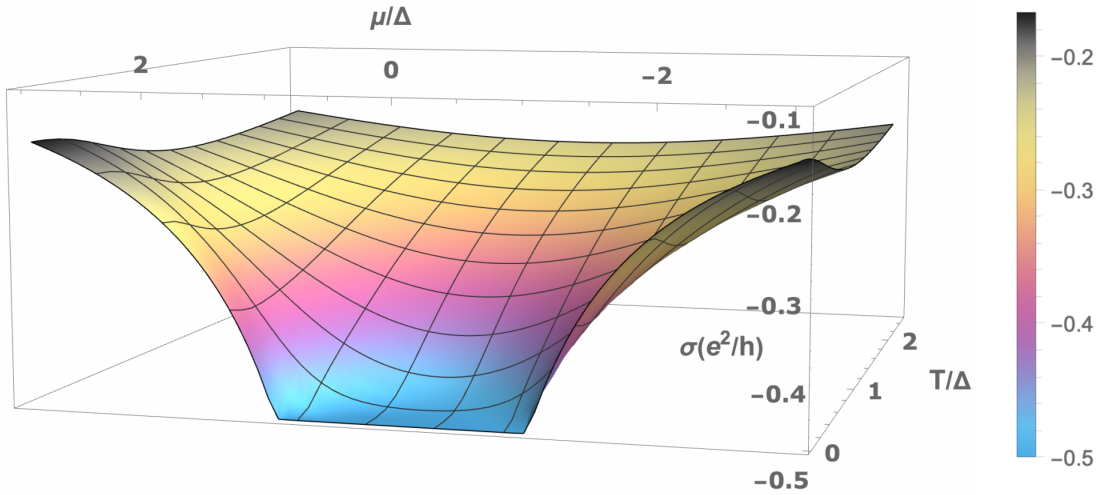


Figure 4.2: Plot of the non-interacting Hall conductivity as a function of (adimensional) temperature and chemical potential, both measured in units of the gap. We can see that at zero temperature, in the range $-\Delta < \mu < \Delta$ the conductivity is zero, as expected.

the temperature measured in units of the gap. The numerical solution of this is given in Figure 4.2, where both the temperature and the gap dependencies are shown.

In the zero temperature limit, the Fermi-Dirac distribution becomes a step function, such that the response function when the chemical potential is in the gap, $\mu = 0$ gives $\chi^{\mathcal{T}(0)}_{j_x j_y} \to -\frac{\omega}{4\pi}$, for $\Lambda \to \infty$. This recovers our result from Dumitriu-Iovanescu (2021), proving that the Hall conductivity is indeed quantised in the right units $\sigma_{\mathrm{H}} = -\frac{1}{2}\frac{e^2}{h}$.

The reason why this is a half-integer is that we only considered one valley, while the physical Hall conductivity which is known to be quantised is the sum of the two (Sinitsyn et al., 2006). The corresponding Chern number depends on the chosen model, as in a hexagonal lattice with time-reversal symmetry $C_1 = 0$, but in the Haldane model $C_1 = -1$.

5 First-order corrections

Among the six first-order Feynman diagrams given in Figure 3.2, only the first three contribute to the correction of the Hall conductivity, since in the limit $\boldsymbol{q} = 0$ the interaction line in the other three carries zero momentum and therefore only contribute to the ground state average of $\langle \hat{j}_{\mathrm{x}} \rangle \langle \hat{j}_{\mathrm{y}} \rangle$ (Giuliani, Vignale, 2005). In the following sections we will calculate each of them individually and discuss the various problems that might arise from only considering first-order diagrams.

5.1 Exchange diagram

For the exchange diagram shown in Figure 4.1(b), we apply the Feynman rules and get:

$$\chi^{\mathcal{T}(\mathrm{EX})}_{j_{\mathrm{x}} j_{\mathrm{y}}}(\boldsymbol{q}, i\omega) = v_{\mathrm{F}}^2 \int \frac{\mathrm{d}^2\boldsymbol{k}}{(2\pi)^2} \int \frac{\mathrm{d}^2\boldsymbol{k}'}{(2\pi)^2} V(\boldsymbol{k}-\boldsymbol{k}') \frac{1}{(\hbar\beta)^2} \sum_{\omega_n, \omega_n'} \mathrm{Tr}\Big[G^{(0)}(\boldsymbol{k}+\boldsymbol{q}, i\omega_n + i\omega)$$
$$G^{(0)}(\boldsymbol{k}'+\boldsymbol{q}, i\omega_n' + i\omega)\, \sigma_y\, G^{(0)}(\boldsymbol{k}', i\omega_n') G^{(0)}(\boldsymbol{k}, i\omega_n)\, \sigma_x \Big], \quad (5.1)$$

which we evaluate in a similar manner to the non-interacting response function discussed above. We can see that by setting $\boldsymbol{q} = 0$ and splitting the terms with $\boldsymbol{k}$ and $\boldsymbol{k}'$ we obtain two fractions which can be integrated individually. We calculate the one in $\boldsymbol{k}$ by performing the angular integration and then summing over the Matsubara frequencies, which lets us express the second fraction from the first by replacing $\sigma_x \leftrightarrow \sigma_y$ and $\boldsymbol{k} \leftrightarrow \boldsymbol{k}'$. When taking the trace of their product in the limit of small ω, we get:

$$\chi^{\mathcal{T}(\mathrm{EX})}_{j_\mathrm{x} j_\mathrm{y}}(i\omega) = \frac{V_c v_\mathrm{F}^2 \hbar\omega\Delta}{32\pi^2} \int_0^\Lambda \mathrm{d}k \int_0^\Lambda \mathrm{d}k' \frac{k\,k' \left(\varepsilon_{\boldsymbol{k}}^2 + \varepsilon_{\boldsymbol{k}'}^2 + 2\Delta^2\right)}{\varepsilon_{\boldsymbol{k}}^3\ \varepsilon_{\boldsymbol{k}'}^3} \left(n^+ - n^-\right)\left(n'^+ - n'^-\right), \quad (5.2)$$

where we considered short range interactions, modelled as a constant contact potential V_c.

It is more convenient to work with adimensional quantities, such that by making the global substitution $x = \beta\varepsilon_k = \beta\sqrt{\hbar^2 v_\mathrm{F}^2 k^2 + \Delta^2}$, we can measure the temperature and the chemical potential in units of the gap $T = 1/(\beta\Delta)$ and $\mu = \mu_0/\Delta$, respectively. This gives:

$$\chi^{\mathcal{T}(\mathrm{EX})}_{j_\mathrm{x} j_\mathrm{y}}(i\omega) = \frac{V_c\,\hbar\omega\,\Delta}{16\pi^2\hbar^4 v_\mathrm{F}^2} \left[\int_{1/T}^{E_\Lambda} \mathrm{d}x' (n'^+ - n'^-) \int_{1/T}^{E_\Lambda} \frac{\mathrm{d}x}{x^2} (n^+ - n^-) + \left(\frac{1}{T} \int_{1/T}^{E_\Lambda} \frac{\mathrm{d}x}{x^2} (n^+ - n^-) \right)^2 \right], \quad (5.3)$$

where E_Λ is an adimensional UV cut-off equal to $\beta\varepsilon_\Lambda = \beta\sqrt{(\hbar v_\mathrm{F})^2\,\Lambda^2 + \Delta^2}$.

5.2 Self-energy diagrams

Unlike the vertex correction (the exchange diagram), calculating the self-energy contributions straight from the two diagrams in Figure 4.1 using the Feynman rules proves to be very problematic. In both of them, the two solid lines before and after the interaction give rise to higher order poles, because their corresponding Green's functions carry the same wave vector and frequency. Since the correlation function is expected to only have simple poles (Giuliani, Vignale, 2005), we conclude that the contributions this method introduces are artifacts of perturbation theory, which are unphysical.

However, an alternative method of carrying out the same calculation can be employed, by extracting our diagrams from the expansion of a new object, which we are going to construct using Green's functions "dressed" with interaction insertions.

5.2.1 Fock self-energy insertion

Since both diagrams (c) and (d) from Figure 4.1 have similar interactions, it is useful to study the self-energy insertion from Figure 5.1 separately, since we will use it later on.

Although there also exists a Hartree self-energy insertion, which can be seen in the last two diagrams from Figure 3.2, it does not contribute to our correlation function since its interaction line carries zero momentum. For this reason we can denote the Fock self-energy at first-order $\Sigma^{(1)}$ and simply refer to it as the "self-energy". It is given by:

$$\Sigma^{(1)}(\boldsymbol{k}, i\omega_n) = -\int \frac{\mathrm{d}^2\boldsymbol{k}'}{(2\pi)^2} \frac{1}{\hbar\beta} \sum_{\omega_n'} V(\boldsymbol{k} - \boldsymbol{k}')\, G^{(0)}(\boldsymbol{k}', i\omega_n'), \quad (5.4)$$

such that after using the Green's function from Equation (4.1) for the case of contact interaction $V(\boldsymbol{k} - \boldsymbol{k}') = V_c$, we obtain a term in the numerator proportional to $\boldsymbol{k}' \cdot \boldsymbol{\sigma}$ which

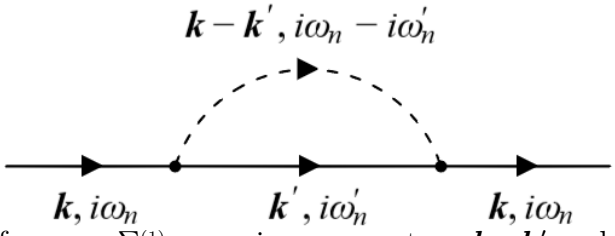


Figure 5.1: Fock self-energy $\Sigma^{(1)}$, carrying momentum $\boldsymbol{k}-\boldsymbol{k}'$ and frequency $i\omega_n - i\omega_n'$.

cancels when we integrate over the angle. We are therefore left with a fraction with poles at $z = \pm\varepsilon_{\boldsymbol{k}'} - \mu$, which can be split into one contribution multiplying $\mathbb{1}_2$ and the other σ_z.

Working once again with the adimensional quantities $x = \beta\varepsilon_{\boldsymbol{k}}$, $T = 1/(\beta\Delta)$ and $E_\Lambda = \beta\varepsilon_\Lambda$, we obtain the following analytic expression for the Fock self-energy:

$$\Sigma^{(1)} = -\frac{V_c\,\Delta^2\,T^2}{4\pi(\hbar v_{\mathrm{F}})^2}\left[\mathbb{1}_2 \underbrace{\int_{1/T}^{E_\Lambda} \mathrm{d}x\, x\left(n^+ + n^-\right)}_{} + \sigma_z \underbrace{\frac{1}{T}\int_{1/T}^{E_\Lambda} \mathrm{d}x\left(n^+ - n^-\right)}_{}\right], \tag{5.5}$$

which is a constant[13], since it does not depend on the external wave vector or frequency when the interaction is modelled by a contact potential. This allows us to express it as:

$$\Sigma^{(1)} = -\frac{V_c\,\Delta^2\,T^2}{4\pi(\hbar v_{\mathrm{F}})^2}\left(\delta\widetilde{\mu}\,\mathbb{1}_2 + \delta\widetilde{\Delta}\,\sigma_z\right) = \delta\mu\,\mathbb{1}_2 + \delta\Delta\,\sigma_z, \tag{5.6}$$

where $\delta\widetilde{\mu}$ and $\delta\widetilde{\Delta}$ are the terms **emphasised** in the Equation 5.5, while in $\delta\mu$ and $\delta\Delta$ we absorbed the overall constant. Our choice of name will only become evident by the end of this section, after introducing dressed Green's functions and Dyson's equation.

5.2.2 Dressed Green's functions

Although working with non-interacting Green's functions seemed to suffice until now, we need to build a more general approach to incorporate interactions in our formalism. By "dressing" the bare solid line with an arbitrary complicated interaction process Σ, we can generate an infinite series covering the whole set of possible interactions that can take place between the end points of the line, provided we choose this process carefully.

We define this new building block Σ to be the sum of the interaction insertions of all orders, that cannot be separated into two by cutting one internal solid line[14](Giuliani, Vignale, 2005). This now enables us to express the dressed Green's function recurrently, as a geometric series, which is illustrated schematically in Figure 5.2. Formally, this recurrence relation is called Dyson's equation and is written as:

$$G = G^{(0)} + G^{(0)}\hbar^{-1}\Sigma\,G^{(0)} + G^{(0)}\hbar^{-1}\Sigma\,G^{(0)}\hbar^{-1}\Sigma\,G^{(0)} + \ldots = G^{(0)} + G^{(0)}\hbar^{-1}\Sigma\,G, \tag{5.7}$$

which can be inverted in matrix form to give $[G\,]^{-1} = [G^{(0)}]^{-1} - \hbar^{-1}\Sigma$. Since our goal is to extract the two Fock self-energies shown on the bottom on Figure 4.1, we can choose $\Sigma^{(1)}$ to be the self-energy that dresses our Green's function. Plugging its simplified form from Equation (5.6) and the expression of the Hamiltonian (2.2) into our solution, we get:

$$G(\boldsymbol{k}, i\omega_n) = \left[(i\hbar\omega_n + \mu - \delta\mu)\mathbb{1}_2 - \hbar v_{\mathrm{F}}\boldsymbol{k}\cdot\boldsymbol{\sigma} - (\Delta + \delta\Delta)\sigma_z\right]^{-1}, \tag{5.8}$$

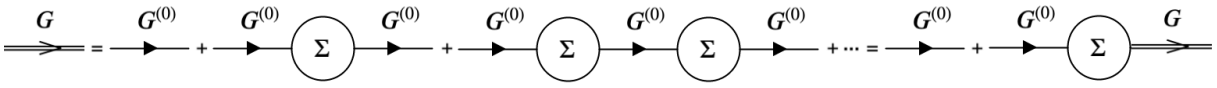


Figure 5.2: Diagrammatic series showing how the dressed Green's function (double line) can be obtained from the non-interacting one (simple line) and the irreducible self-energy.

which clearly resembles our result in Equation (4.1). Comparing the two, we can see that going from non-interacting to dressed Green's functions effectively changes the chemical potential $\mu \to \mu - \delta\mu$ and the gap $\Delta \to \Delta + \delta\Delta$.

5.2.3 Renormalisation

Seeing that by substituting $\mu \to \mu - \delta\mu$ and $\Delta \to \Delta + \delta\Delta$ we keep our non-interacting response function intact, we are now interested in calculating these two corrections to the chemical potential and the gap, in order to see their explicit temperature dependence.

Although we can analytically evaluate the integrals in Equation 5.5, they both contain divergences in the limit E_Λ, which we need to discard. For the term shifting μ we get:

$$\begin{aligned}\delta\widetilde{\mu} \;=& \frac{1}{2T^2} - \frac{1}{T}\ln\left(\frac{e^{(1+\mu)/T}+1}{e^{(1-\mu)/T}+1}\right) + \mathrm{Li}_2(-e^{(1-\mu)/T}) - \mathrm{Li}_2(-e^{(1+\mu)/T}) \\ &+ \underbrace{\frac{E_\Lambda^2}{2} - E_\Lambda \ln\left(\frac{e^{E_\Lambda-\mu/T}+1}{e^{E_\Lambda+\mu/T}+1}\right) + \mathrm{Li}_2(-e^{E_\Lambda+\mu/T}) - \mathrm{Li}_2(-e^{E_\Lambda-\mu/T})}, \end{aligned} \tag{5.9}$$

where Li_2 is called a dilogarithm[15], while for the one changing the gap:

$$\delta\widetilde{\Delta} = \frac{1}{T}\Big[-\frac{1}{T} + \ln\left((e^{(1-\mu)/T}+1)(e^{(1+\mu)/T}+1)\right) + \underbrace{E_\Lambda - \ln\left((e^{E_\Lambda-\mu/T}+1)(e^{E_\Lambda+\mu/T}+1)\right)}\Big]. \tag{5.10}$$

The two sequences marked above hide linear and quadratic divergences, which are clearly unphysical. The formal procedure that needs to be employed to amend our approach is called renormalisation, and it comes down to absorbing these infinities into the bare parameters μ and Δ, by arguing this is necessary in order to recover experimental results from our theory. Better explained in Lancaster, Blundell (2014): "Renormalization is not an exercise in hiding infinities, it's an exercise in making a theory describe real life".

After removing all the divergences from the parameters $\delta\widetilde{\mu}$ and $\delta\widetilde{\Delta}$, from here on we will only consider their finite parts when referring to them.

5.2.4 Zero-order diagram with dressed Green's functions

Looking back at Figure 4.1, it becomes clear now that by dressing the two solid lines in diagram (a) with the Fock self-energy insertion, both (c) and (d) are obtained, provided we make sure we only include contributions in interactions of first-order. To ensure it, we multiply $\Sigma^{(1)}$ by a coupling constant g, in which we perform the series expansion.

[13] Notice that we drop the $\boldsymbol{k}$ and $i\omega_n$ dependence from here on.
[14] This is what we call "irreducible" and is a paramount condition to ensure that we avoid over-counting.
[15] It is a special case of the polylogarithm Li_n and is defined by the integral $\mathrm{Li}_2(z) = \int_z^0 \mathrm{d}x \ln(-x)/x$.

By substituting $\mu \to \mu - g\,\delta\mu$ and $\Delta \to \Delta + g\delta\,\Delta$ into the zero-th order correlation function given in Equation (4.3), we obtain the following form:

$$\chi^{\mathcal{T}(\Sigma)}_{j_\mathrm{x}j_\mathrm{y}}(i\omega) = \frac{v_\mathrm{F}^2\hbar\omega(\Delta + g\,\delta\Delta)}{4\pi}\int_0^\Lambda \mathrm{d}k\,k\frac{n_\mathrm{F}(\varepsilon^*_{\boldsymbol{k}} - \mu + g\,\delta\mu) - n_\mathrm{F}(-\varepsilon^*_{\boldsymbol{k}} - \mu - g\,\delta\mu)}{(\varepsilon^*_{\boldsymbol{k}})^3}, \quad (5.11)$$

where $\varepsilon^*_{\boldsymbol{k}} = \sqrt{(\hbar v_\mathrm{F})^2k^2 + (\Delta + g\,\delta\Delta)^2}$ also depends on the coupling constant. We then expand this in powers of g, only keeping the first-order terms, and work once again with adimensional quantities. The final form of the self-energy contribution is given by:

$$\chi^{\mathcal{T}(\mathrm{SE})} = -\frac{V_c\,\hbar\omega\Delta}{16\pi^2\hbar^4v_\mathrm{F}^2}\int_{1/T}^{E_\Lambda}\mathrm{d}x\left[\frac{\delta\widetilde{\mu}}{x^2}\frac{\partial}{\partial x}\left(n^+ + n^-\right) + \frac{1}{T}\frac{\delta\widetilde{\Delta}}{x^3}\frac{\partial}{\partial x}\left(n^+ - n^-\right) - \right.$$
$$\left. -\frac{3}{T}\frac{\delta\widetilde{\Delta}}{x^4}\left(n^+ - n^-\right) + \underbrace{T\frac{\delta\widetilde{\Delta}}{x^2}\left(n^+ - n^-\right)}\right], \quad (5.12)$$

where we can see that the highlighted part cancels exactly the first term in the exchange correlation function given in Equation (5.3), since $\delta\widetilde{\Delta} = \frac{1}{T}\int \mathrm{d}x\,(n^+ - n^-)$.

This result is consistent with our previous calculation from Dumitriu-Iovanescu (2021), where working in the zero temperature formalism, we also obtained canceling contributions between the exchange and the self-energy diagrams, plus a remaining diverging term which was not treated as part of any bare parameter renormalisation.

5.3 Results

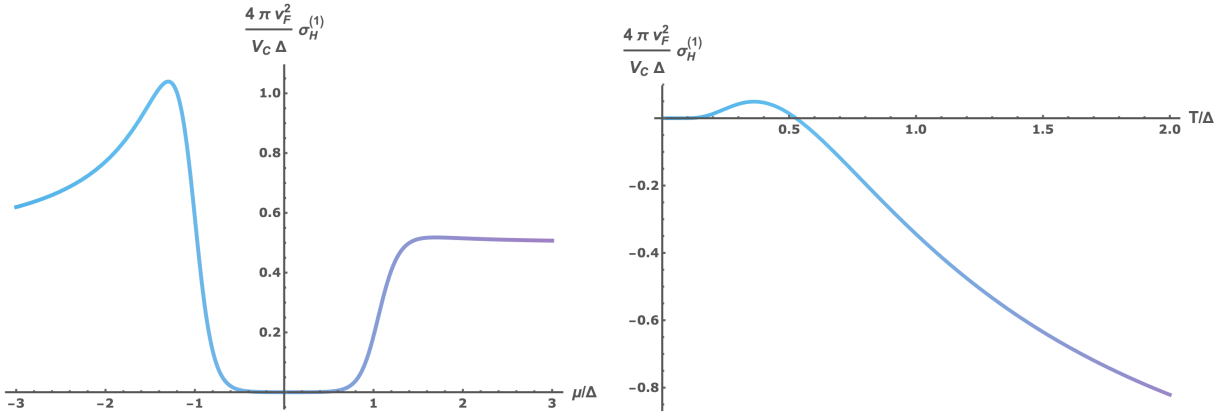
All there is left to do now is to add the contribution from the exchange and the two self-energy diagrams we calculated in the previous sections, to determine the full correlation function. After analytically continuing $\omega \to -i\omega + \eta$ and using the Kubo formula, we get our final result for the first-order correction in the Hall conductivity:

$$\sigma^{(1)}_\mathrm{H}(T,\mu) = \frac{e^2}{2h}\frac{V_c\,\Delta}{4\pi(\hbar v_\mathrm{F})^2}\left[\left(\frac{1}{T}\int_{1/T}^{E_\Lambda}\frac{\mathrm{d}x}{x^2}\left(n^+ - n^-\right)\right)^2 + \delta\widetilde{\Delta}\frac{3}{T}\int_{1/T}^{E_\Lambda}\frac{\mathrm{d}x}{x^4}\left(n^+ - n^-\right) - \right.$$
$$\left. -\delta\widetilde{\Delta}\,\frac{1}{T}\int_{1/T}^{E_\Lambda}\frac{\mathrm{d}x}{x^3}\frac{\partial}{\partial x}\left(n^+ - n^-\right) - \delta\widetilde{\mu}\int_{1/T}^{E_\Lambda}\frac{\mathrm{d}x}{x^2}\frac{\partial}{\partial x}\left(n^+ + n^-\right)\right]. \quad (5.13)$$

Since we already removed the divergences in $\delta\widetilde{\mu}$ and $\delta\widetilde{\Delta}$, we know all the terms above are convergent and therefore the integrals can be evaluated numerically. Plotting separately the dependence of $\sigma^{(1)}_\mathrm{H}$ on the temperature for fixed μ - Figure 5.3(a), and on the chemical potential for fixed T - Figure 5.3(b), we can see that the behaviour is atypical.

For a fixed temperature, we would expect the trend of $\sigma^{(1)}_\mathrm{H}$ to be symmetrical for μ and $-\mu$, since they each correspond to the contribution coming from electrons and from holes, respectively. However, seeing that our plot (a) appears to "favour" the holes, we conclude that something in our analysis is either incorrect or incomplete. Similarly, the behaviour of the conductivity in the undoped case ($\mu = 0$) is strange in the region $0 < T < 0.5\Delta$.

A plausible explanation for these results is that our theory is built around Dyson's equation and the idea of constructing Σ as a geometric series, which we truncated at first-order. As a consequence of not including the infinite series of insertions in Σ, we introduced unphysical contributions to our correlation function (Giuliani, Vignale, 2005), which need to be amended by imposing additional constraints.



(a) Fixed $T = 0.01$ (b) Fixed $\mu = 0$

Figure 5.3: Plots showing the first-order correction to the Hall conductivity at fixed temperature (a) and fixed chemical potential (b). The corrections are measured in units of $e^2/2h$ and are rescaled by an overall factor of $V_c\Delta/(4\pi v_{\mathrm{F}}^2)$.

5.3.1 Number of particles conservation

By plotting the exchange separately, it can easily be seen that the unphysical behaviour comes from the two self-energy diagrams, as we expected. Since our analysis was only possible by introducing additional terms that renormalise the chemical potential and the gap, it suggests that the contribution coming from these terms might bypass a conservation law. Both terms on the second line of Equation (5.13) contain derivatives of the Fermi-Dirac occupation functions, which hints that we should check whether the number of particles in our system in conserved.

Initially, this is given by $N = \int \frac{\mathrm{d}^2\boldsymbol{k}}{(2\pi)^2}(n^+ + n^-)$. One way to ensure N remains unchanged when introducing $\delta\mu$ and $\delta\Delta$ is by differentiating it with respect to the coupling constant and setting the result to zero: $\partial N^*/\partial g = 0$. This is equivalent to saying that the number of particles added by the interactions is null, where we use N^*:

$$N^* = \int \frac{\mathrm{d}^2\boldsymbol{k}}{(2\pi)^2}\Big[n_{\mathrm{F}}(\varepsilon^*_{\boldsymbol{k}} - \mu + g\,\delta\mu) + n_{\mathrm{F}}(-\varepsilon^*_{\boldsymbol{k}} - \mu + g\,\delta\mu)\Big], \tag{5.14}$$

to differentiate it from the non-interacting one, and $\varepsilon^*_{\boldsymbol{k}} = \sqrt{(\hbar v_{\mathrm{F}})^2k^2 + (\Delta + g\delta\,\Delta)^2}$ as before.

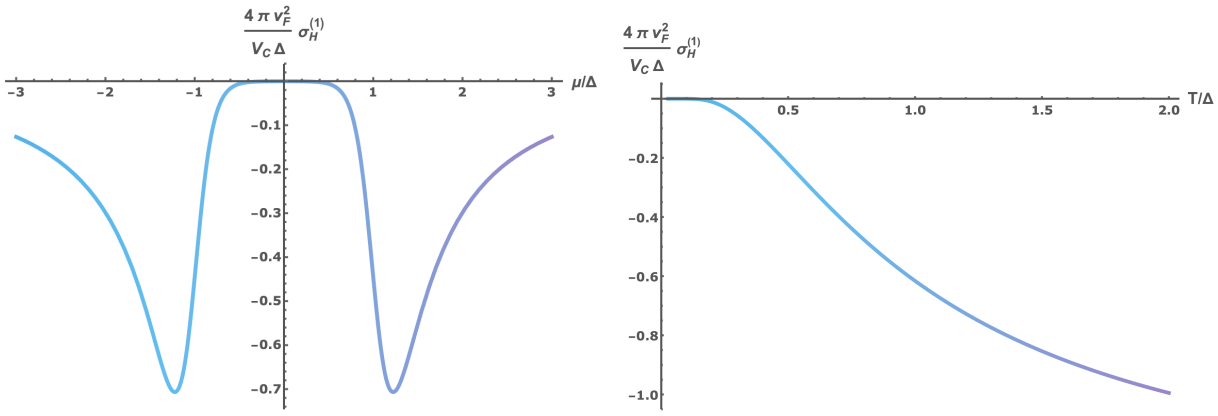
After performing this calculation we obtain the following relation:

$$\delta\mu \int_{1/T}^{E_\Lambda} \mathrm{d}x\, x\, \frac{\partial}{\partial x}(n^+ - n^-) + \delta\Delta\, \frac{1}{T} \int_{1/T}^{E_\Lambda} \mathrm{d}x \frac{\partial}{\partial x}(n^+ + n^-) = 0, \tag{5.15}$$

which allows us to constrain $\delta\mu$[16] as a function of $\delta\Delta$. We notice that the terms appearing in our relation above contain the derivatives of the occupation function, which is something we expect since similar terms were causing the atypical behaviour of $\sigma_{\mathrm{H}}^{(1)}$ before.

The plots we obtain using this method for the first-order correction to the Hall con-

[16] We choose to redefine $\delta\mu$ in terms of $\delta\Delta$ and not the other way around, on the grounds that the change in the chemical potential, and not the gap, is what physically introduces new particles.



(a) Fixed $T = 0.01$ (b) Fixed $\mu = 0$

Figure 5.4: Plots showing the first-order correction to the Hall conductivity at fixed temperature (a) and fixed chemical potential (b), after imposing that the number of particles is conserved. The corrections are measured in units of $e^2/2h$ and are rescaled by an overall factor of $V_c\,\Delta/(4\pi v_F^2)$, the same as before.

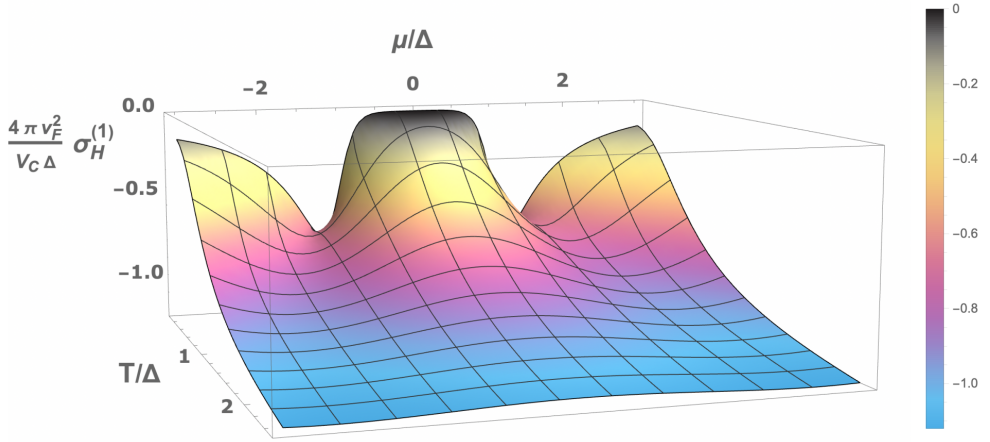


Figure 5.5: Surface plot of the first-order correction to the Hall conductivity as a function of (adimensional) temperature and chemical potential, both measured in units of the gap.

ductivity are shown in Figure 5.4 and allow us to compare our result with our previous one. Additionally, a surface plot showing both the temperature and the chemical potential dependencies is given in Figure 5.5.

5.3.2 Discussion

After imposing that the number of particles is conserved, we can see that the trend in our first-order correction to the conductivity changes in both cases (a) and (b). At fixed temperature, the correction is symmetrical in $\pm\mu$, and it goes to zero in the limit of large (positive or negative) doping. Although at small temperatures we would expect it to be nearly zero in the range $-\Delta < \mu < \Delta$, numerical integration prevents us from taking $T = 0$, such that the effects of finite temperature are always present, as can be seen from the sharp variation as μ approaches $\pm\Delta$. At fixed chemical potential, the correction is always

negative and its modulus increases with increasing temperature, eventually converging to:

$$\sigma_{\mathrm{H}}^{(1)} \to -\frac{V_c\,\Delta}{4\pi(\hbar v_{\mathrm{F}})^2}\,(4\ln 2)\,\frac{e^2}{2h}. \tag{5.16}$$

Comparing our results with the literature, Coleman, Hill (1985) states that higher-order corrections to the topological Hall conductivity should vanish at all orders, and therefore also in our case for $\mu = 0$. Although this is true in the limit $T \to 0$, as shown in Figures 5.4 and 5.5, this conclusion cannot be generalised for $T \neq 0$, due to the ambiguity of defining the topological part of the Hall conductivity at finite temperature. A short discussion concerning this topic can be found in Appendix C.

Looking back at the details concerning our contact potential from Section 2.3, we can evaluate the constant term from Equation 5.16 in terms of the interaction strength λ:

$$\frac{V_c\,\Delta}{4\pi(\hbar v_{\mathrm{F}})^2} = \lambda\,\underbrace{\frac{e^2}{4\pi\varepsilon_0\hbar v_{\mathrm{F}}}}\,\frac{\Delta}{\hbar v_{\mathrm{F}}\Lambda} = \lambda\,\alpha\frac{\Delta}{\hbar v_{\mathrm{F}}\Lambda} \approx \lambda\,\alpha\frac{\Delta}{\varepsilon_\Lambda}, \tag{5.17}$$

where the emphasised term is the fine-structure constant α and Λ is the natural cut-off in momentum. We therefore conclude that the order of magnitude of our correction depends linearly on the strength of the electron-electron interaction, the fine structure of our material and the ratio $\Delta/\varepsilon_\Lambda$, which can be tuned by changing the gap opening.

6 Conclusion

The aim of this project was to calculate first-order corrections to the anomalous Hall conductivity of massive Dirac fermions, for short-range interactions. To avoid dealing with numerous IV divergences, these interactions were approximated by a contact potential, which does not depend on the momentum carried along the interaction line.

The finite temperature formalism was introduced to go beyond the zero temperature one carried out last semester, and therefore generalise our results to more realistic scenarios. Matsubara Green's functions enabled us to perform diagrammatic perturbation theory on the current-current correlation function, which we calculated at zero and first-order. The result for the non-interacting case recovers the one we obtained last semester, and it is also consistent with the literature. The dependence of the Hall conductivity on the temperature and the gap was illustrated in a surface plot, which was later used as a subject of discussion in Appendix C, for topological considerations at finite temperature.

First-order vertex and self-energy corrections were calculated using Dyson's equation, but as a consequence of truncating an infinite geometric series, the result revealed atypical behaviour. This was amended by imposing that the number of particles in the system is conserved, which ultimately gave rise to corrections that vary with temperature and the chemical potential. These were displayed as distinct plots, as well as a surface plot showing both dependencies simultaneously.

In the limit of large temperature, a converging first-order correction was found. Since it depends on the strength of the interaction, the fine-structure constant of the material and the ratio between the gap and some cut-off energy, its value can be tuned.

Although for a weakly interacting system we can argue that this is the dominant correction to the anomalous Hall conductivity, this is not true when interactions are strong and perturbation theory fails. In the latter case, we need to employ a different approach which accounts for infinite terms in the series (Giuliani, Vignale, 2005).

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A Feynman rules

For the first-order diagrams shown in Figure 3.2, the following Feynman Rules apply:

Step 1: Draw the six topologically distinct connected diagrams formed from two external vertices (in our case j^x and j^y), one interaction line, and four oriented solid lines.

Step 2: Associate to each solid line a wave vector $\boldsymbol{k}_n$ and a fermionic Matsubara frequency $i\omega_n$, and to the interaction line a wave vector $\boldsymbol{k}_m$ and a bosonic frequency $i\omega_m$.

Step 3: Conserve both the wave vector and the frequency at each vertex, including a wave vector $\pm\boldsymbol{q}$ and a bosonic frequencies $\pm i\omega$ at the two vertices, respectively.

Step 4: Associate to each solid line a non-interacting Matsubara Green's function $G^{(0)}_{\alpha\beta}(i\omega_n)$, given in Equation (3.14), to the interaction line a factor $V(\boldsymbol{k}_m)$, and to each of the two external vertices a matrix element (in our case, $j^{\mathrm{x}}_{\alpha\beta}$ and $j^{\mathrm{y}}_{\gamma\delta}$).

Step 5: Sum over the two internal fermionic Matsubara frequencies[17] (which are left after the frequency conservation), integrate over the two wave vectors[18] (which are left after the momentum conservation) and sum over the internal state labels.

Step 6: To ensure the contour is closed in the upper half plane, the sum over the Matsubara frecquencies should be interpreted as $\lim_{\eta\to 0^+} \frac{1}{\hbar\beta}\sum_n G^{(0)}(i\omega_n)e^{i\eta\omega_n}$.

Step 7: Multiply by a factor $(-\frac{1}{\hbar})(-1)^{\mathrm{F}}$, where F is the number of closed fermion loops.

Step 8: Perform the analytic continuation for the external frequency $i\omega \to \omega + i\eta$ to retrieve the retarded response function from the Matsubara correlation function.

Similarly, when calculating the non-interacting Hall current-current response function from the zero-th order diagram, the following rules apply:

Step 1: Associate wave vectors $\boldsymbol{k}$ and $\boldsymbol{k}'$ and fermionic Matsubara frequencies $i\omega_n$ and $i\omega'_n$ to the two solid lines.

Step 2: After conserving the momentum and the frequency at each vertex, associate non-interacting Matsubara Green's functions $G^{(0)}(\boldsymbol{k}+\boldsymbol{q}, i\omega_n + i\omega)$ and $G^{(0)}(\boldsymbol{k}, i\omega_n)$ to the two solid lines and matrix elements $j^{\mathrm{x}}_{\alpha\beta}$ and $j^{\mathrm{y}}_{\gamma\delta}$ to the vertices.

Step 3: Sum over the Matsubara frequencies $i\omega_n$ ensuring the contour is closed in the upper half-plane, integrate over the wave vector $\boldsymbol{k}$ and sum over the internal state.

Step 4: Multiply by an overall factor $-\frac{1}{\hbar}$.

Step 5: Perform the analytic continuation for the external frequency $i\omega \to \omega + i\eta$.

These sets of rules are paraphrased from Section 6.4. in Giuliani, Vignale (2005).

[17] This sum will take the form $\frac{1}{\hbar\beta}\sum_n \ldots$, where n is a label used for the fermionic Matsubara frequencies.

[18] This integral will take the form $\int \frac{\mathrm{d}\boldsymbol{k}^2}{(2\pi)^2} \ldots$, where $\boldsymbol{k}$ is a 2-dimensional wave vector.

B Analytic continuation

In the eigenstate representation of the non-interacting Hamiltonian $\{|\psi_n\rangle\}$, also called the Lehmann representation, the retarded response function is given by:

$$\begin{aligned}
\chi^{\mathrm{R}}_{\mathrm{AB}}(\tau) &= -\frac{i}{\hbar}\,\theta(\tau)\,\langle[\hat{A}(\tau),\hat{B}]_{\mathrm{B/F}}\rangle \\
&= -\frac{i}{\hbar}\,\theta(\tau)\sum_n \frac{1}{\mathcal{Z}} e^{-\beta\hat{H}}\,\langle\psi_n|\left(\hat{A}(\tau)\hat{B}\mp\hat{B}\hat{A}(\tau)\right)|\psi_n\rangle \\
&= -\frac{i}{\hbar}\,\theta(\tau)\sum_n \frac{1}{\mathcal{Z}} e^{-\beta\hat{H}}\,\langle\psi_n|\left(e^{i\hat{H}\tau/\hbar}\hat{A}\,e^{-i\hat{H}\tau/\hbar}\hat{B}\mp\hat{B}\,e^{i\hat{H}\tau}\hat{A}\,e^{-i\hat{H}\tau/\hbar}\right)|\psi_n\rangle \\
&= -\frac{i}{\hbar}\,\theta(\tau)\sum_n \frac{1}{\mathcal{Z}} e^{-\beta\varepsilon_n} e^{i\varepsilon_n\tau/\hbar}\,\langle\psi_n|\,\hat{A}\Big(\sum_m |\psi_m\rangle\langle\psi_m|\Big)e^{-i\hat{H}\tau/\hbar}\hat{B}\,|\psi_n\rangle \\
&\quad \pm \frac{i}{\hbar}\,\theta(\tau)\sum_n \frac{1}{\mathcal{Z}} e^{-\beta\varepsilon_n}\, e^{-i\,\varepsilon_n\tau/\hbar}\,\langle\psi_n|\,\hat{B}\Big(\sum_m |\psi_m\rangle\langle\psi_m|\Big)e^{i\hat{H}\tau/\hbar}\hat{A}\,|\psi_n\rangle \\
&= -\frac{i}{\hbar}\,\theta(\tau)\sum_{n,m} \frac{1}{\mathcal{Z}} e^{-\beta\varepsilon_n}\left(e^{i(\varepsilon_n-\varepsilon_m)\tau/\hbar}\,A_{nm}B_{mn}\mp e^{-i\,(\varepsilon_n-\varepsilon_m)\tau/\hbar}\,B_{nm}A_{mn}\right) \\
&= -\frac{i}{\hbar}\,\theta(\tau)\sum_{n,m} \frac{1}{\mathcal{Z}} A_{nm}B_{mn} e^{i(\varepsilon_n-\varepsilon_m)\tau/\hbar}\left(e^{-\beta\varepsilon_n}\mp e^{-\beta\varepsilon_m}\right), \qquad \text{(B.1)}
\end{aligned}$$

where $A_{nm}=\langle\psi_n|\,\hat{A}\,|\psi_m\rangle$, $B_{mn}=\langle\psi_m|\,\hat{B}\,|\psi_n\rangle$ and we relabelled $n\leftrightarrow m$ for the last step. In Fourier space[19] it becomes:

$$\begin{aligned}
\chi^{\mathrm{R}}_{\mathrm{AB}}(\omega) &= \int_{-\infty}^{\infty}\chi^{\mathrm{R}}_{\mathrm{AB}}(\tau)\,e^{i(\omega+i\eta)\tau}d\tau \\
&= -\frac{i}{\hbar}\frac{1}{\mathcal{Z}}\sum_{n,m} A_{nm}B_{mn}\left(e^{-\beta\varepsilon_n}\mp e^{-\beta\varepsilon_m}\right)\int_0^{\infty} e^{i(\varepsilon_n-\varepsilon_m)\tau/\hbar}\,e^{i(\omega+i\eta)\tau}d\tau \\
&= -\frac{\not{i}}{\hbar}\frac{1}{\mathcal{Z}}\sum_{n,m}\frac{(-1)A_{nm}B_{mn}\left(e^{-\beta\varepsilon_n}\mp e^{-\beta\varepsilon_m}\right)}{\not{i}\left((\varepsilon_n-\varepsilon_m)/\hbar+\omega+i\eta\right)} \\
&= \frac{1}{\mathcal{Z}}\sum_{n,m}\frac{A_{nm}B_{mn}}{\varepsilon_n-\varepsilon_m+\hbar\omega+i\hbar\eta}\left(e^{-\beta\varepsilon_n}\mp e^{-\beta\varepsilon_m}\right). \qquad \text{(B.2)}
\end{aligned}$$

Similarily, the Matsubara correlation function is given by:

$$\begin{aligned}
\chi_{\mathrm{AB}}(\tau) &= -\frac{1}{\hbar}\sum_n\frac{1}{\mathcal{Z}}\,\langle\psi_n|\left(e^{-\beta\hat{H}}\hat{A}(\tau)\hat{B}\right)|\psi_n\rangle \\
&= -\frac{1}{\hbar}\sum_n\frac{1}{\mathcal{Z}}\,\langle\psi_n|\left(e^{-\beta\hat{H}}e^{\hat{H}\tau/\hbar}\hat{A}\,e^{-\hat{H}\tau/\hbar}\hat{B}\right)|\psi_n\rangle \\
&= -\frac{1}{\hbar}\sum_n\frac{1}{\mathcal{Z}}e^{-\beta\varepsilon_n}e^{\varepsilon_n\tau/\hbar}\,\langle\psi_n|\,\hat{A}\Big(\sum_m|\psi_m\rangle\langle\psi_m|\Big)\,e^{-\hat{H}\tau/\hbar}\hat{B}\,|\psi_n\rangle \\
&= -\frac{1}{\hbar}\sum_{n,m}\frac{1}{\mathcal{Z}}e^{-\beta\varepsilon_n}e^{(\varepsilon_n-\varepsilon_m)\tau/\hbar}A_{nm}B_{mn}, \qquad \text{(B.3)}
\end{aligned}$$

[19] Note that the exponential term in the Fourier transform has an additional factor of $+i\eta$, specific to the retarded response function. This ensures that the θ-function is recovered and causality is preserved.

which in Fourier:

$$\begin{aligned}
\chi_{\text{AB}}(\omega) &= \int_0^{\hbar\beta} \langle[\hat{A}(\tau), \hat{B}(0)]\rangle_0 \; e^{i\omega\tau} d\tau \\
&= -\frac{1}{\hbar}\int_0^{\hbar\beta} \sum_{n,m} \frac{1}{\mathcal{Z}} e^{-\beta\varepsilon_n} e^{(\varepsilon_n-\varepsilon_m)\tau/\hbar} \, A_{nm}B_{mn} \, e^{i\omega\tau} d\tau \\
&= -\frac{1}{\hbar}\sum_{n,m} \frac{1}{\mathcal{Z}} A_{nm}B_{mn} e^{-\beta\varepsilon_n} \int_0^{\hbar\beta} e^{\tau((\varepsilon_n-\varepsilon_m)/\hbar+i\omega)} d\tau \\
&= -\frac{1}{\hbar}\sum_{n,m} \frac{1}{\mathcal{Z}} A_{nm}B_{mn} e^{-\beta\varepsilon_n} \frac{e^{\beta(\varepsilon_n-\varepsilon_m+i\hbar\omega)}-1}{(\varepsilon_n-\varepsilon_m)/\hbar+i\omega} \\
&= \frac{1}{\mathcal{Z}}\sum_{n,m} \frac{A_{nm}B_{mn}}{\varepsilon_n-\varepsilon_m+i\hbar\omega}\left(e^{-\beta\varepsilon_n} - e^{-\beta\varepsilon_m}e^{i\beta\hbar\omega}\right) \\
&= \frac{1}{\mathcal{Z}}\sum_{n,m} \frac{A_{nm}B_{mn}}{\varepsilon_n-\varepsilon_m+i\hbar\omega}\left(e^{-\beta\varepsilon_n} \mp e^{-\beta\varepsilon_m}\right).
\end{aligned} \tag{B.4}$$

We can use the conclusion reached in Chapter 2, Section 7 from Brown, Churchill (2009), which states that if two functions (both analytic on a domain $\mathcal{D}$) coincide on a subdomain or even on a line segment lying in that domain, then they are equal to each other everywhere on $\mathcal{D}$. This statement is enough to prove that the retarded response and the Matsubara correlation function are special cases of the same function:

$$\chi_{\text{AB}}(z) = \frac{1}{\mathcal{Z}}\sum_{n,m} \frac{A_{nm}B_{mn}}{\varepsilon_n-\varepsilon_m+z}\left(e^{-\beta\varepsilon_n} \mp e^{-\beta\varepsilon_m}\right), \tag{B.5}$$

which allows us to use the analytic continuation procedure discussed in Section 3.5. This is shown in Figure B.1. Note that the new function χ_{AB} is analytical everywhere expect on the real axis, due to the poles at $\varepsilon_n = \varepsilon_m$.

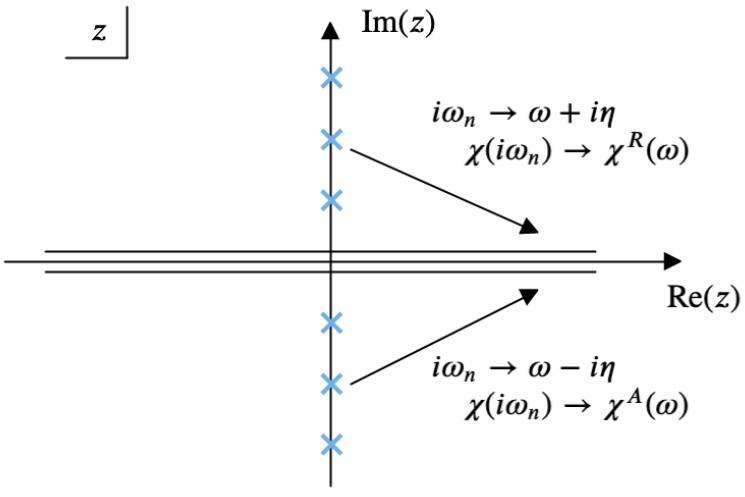


Figure B.1: The analytic continuation procedure obtained by sending the Matsubara frequencies $\omega_n \to \omega \pm i\eta$, such that the correlation function goes to the retarded or the advanced response functions, respectively. The lines parallel to the real axis emphasise that the purely real arguments z are discarded, ensuring that $g(z)$ is analytic everywhere.

C Topological aspects at finite temperature

We saw that the way we relate the Hall conductivity to the Berry curvature and the Chern number is restricted to integrating over the entire Brillouin zone. At zero temperature, this can be done by placing the Fermi level inside the gap, such that all the electrons occupy the states in the lower band.

At zero temperature, any contribution of the doping would be added on top of this, such that the integrals of the occupation functions could always be separated to account for the two distinct contributions. However, at finite temperature the problem changes radically: even when $\mu = 0$, the particles in the lower band have a finite probability to occupy a state in the upper band, give by the Fermi-Dirac distribution. This does not allow us to separate the contribution of a hypothetically full band from the rest, and we can therefore not discuss explicitly about topology.

From physical consideration, both a finite doping at zero temperature and a finite temperature with the chemical potential in the gap have a similar effect: to introduce probabilities for particles to occupy states in the upper band. From our result for σ_{H} as a function of both T and μ, we decided to plot the absolute difference of $\sigma_{\text{H}}(0, \mu) - \sigma_{\text{H}}(T, 0)$ and to identify the region, if there is one, in which this difference gives zero.

This region would give a mapping between T and μ, such that we can understand the topology at finite temperature and $\mu = 0$ in terms of the chemical potential that we is needed at $T = 0$ to replicate the same effect. Knowing this μ, the contribution from the upper band can be subtracted from the total Hall conductivity that we have initially, allowing us to separate the topological part.

From Figure C.1, we see that such a region exists, and therefore there is a mapping between T and μ. However, since this was performed at zero-order, we are unsure if this applies to higher order contributions as well, as we need a more detailed analysis to check whether this is physically meaningful.

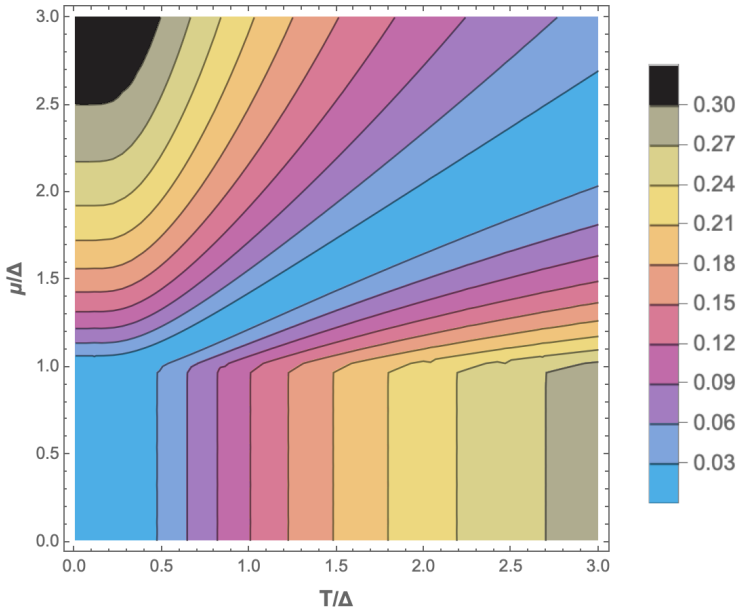


Figure C.1: Contour plot of the absolute difference $\sigma_{\text{H}}(0, \mu) - \sigma_{\text{H}}(T, 0)$.